Do Now · retrieval

RAM, ROM and virtual storage

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why are SSDs and HDDs physically different?

Q2

What purpose does processor cache serve?

Q3

Which is normally volatile: RAM or ROM?

Q4

Sorted array [2,6,10,14,18,22,26,30]. Find 22 with midpoint indexes and range updates.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

RAM versus ROM — RAM supports active execution and is volatile; ROM and firmware storage retain instructions without power.

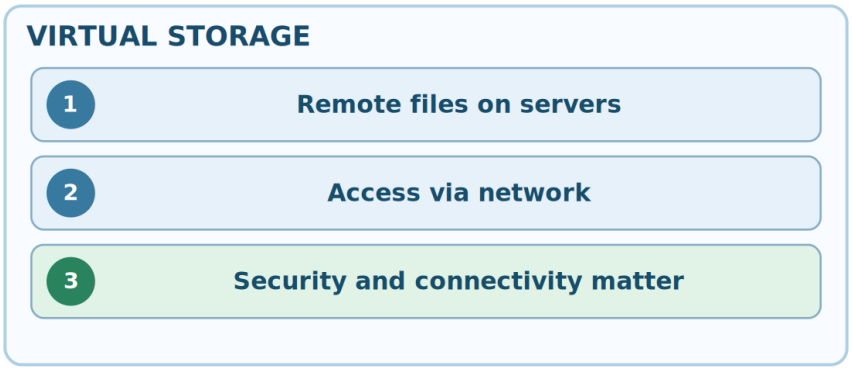
Core idea 2

Define virtual storage — A remote or cloud storage service keeps files on network-accessible systems, potentially available on multiple devices.

Core idea 3

Distinguish virtual memory — OS virtual memory involves memory addressing/paging with secondary storage, taught again in OS memory management.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

RAM — Volatile read/write working memory.

ROM — Non-volatile firmware/boot storage.

virtual storage — Remote network-accessible file storage.

virtual memory — OS technique using address translation and backing storage.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Why can unsaved editing work disappear after power loss while firmware remains?

Follow the worked method

1. Predict — Unsaved temporary program state depends on volatile RAM and can disappear without power.

2. Explain — Firmware persists in non-volatile storage, allowing startup instructions to remain available.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict

Virtual/cloud storage supports access from multiple locations and may offer remote copies.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

State typical use and whether RAM and ROM are volatile.

Question 2

Explain two benefits and one drawback of network-based virtual storage for pupil coursework.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Distinguish virtual storage from virtual memory and explain one limitation of each.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Virtual cloud storage is another name for OS virtual memory.

Use key vocabulary from memory

1. Explain RAM in your own words.

2. Explain ROM in your own words.

3. Explain virtual storage in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.